

Elbow and Forearm

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Michael Simon climbing in Kalymnos, Greece, Photo Archive Michael Simon

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8.1 Overview

Pathologies in the area of the elbow and forearm are the fourth most common type of injury in climbing after finger, shoulder, and hand injuries, accounting for 7.7% of all pathologies in our own patient sample [1]. Furthermore, a study by Gronhaug et al., which included 667 active climbers with complaints around the elbow, found an incidence of 17.7% of all climbing-related injuries, making this the third most common pathology in climbers [2]. Hereby, mainly chronic pathologies caused by overload are found. Epicondylitis humeri, with a share of up to 8.4%, is one of the ten most common diagnoses in climbers. Elbow pain can also be caused by brachialis, biceps, and triceps muscle tendinopathies [3]. Traumatic injuries in the area of the elbow and forearm, such as ruptures of the distal biceps tendon as well as higher-value muscle or vascular injuries, occur occasionally in climbing and require a quick, correct diagnosis and prompt surgical treatment. If load-dependent pain in the forearm persists for a long time after climbing, the differential diagnosis of the rarely occurring functional compartment syndrome of the forearm musculature must also be considered [4]. In the following section, the specifics of the main climbing-related injuries around the elbow, their diagnosis, and their treatment are discussed.

8.2 Lateral and Medial Epicondylitis

Epicondylitis humeri is one of the most common complaints among climbing athletes, although, in recent years, a slight decrease in incidence has been observed in our own patient sample. Nevertheless, with 3.3% of all complaints, it is still among the ten most common climbing-related pathologies [1]. A distinction is made between lateral (radial) and medial (ulnar) epicondylitis, with medial epicondylitis occurring more often in climbers [5, 6]. Patients report local pressure pain at the epicondyle and, sometimes, pain radiating into the forearm muscles, a decrease in strength, and pain at rest. The com-

plaints are caused by repetitive overstrain of the origin of the forearm extensors (lateral side) or flexors (medial side), whereby chronic micro-trauma leads to degenerative changes and ultimately to tendinopathy [7]. Muscular imbalance between the forearm flexors and extensors can also be one of the triggers for epicondylitis, since training the antagonists (extensor group) is often neglected by many climbers. The diagnosis is made by clinical examination, including provocation tests (lateral, isometric extension of the wrist as well as the middle finger when the elbow is extended; medial, isometric flexion in the wrist as well as the fingers when the elbow is extended) and sonography, if necessary. Further diagnostics in the form of X-ray and MRI should only be performed in the case of refractory complaints under conservative therapy to exclude any other underlying pathology (e.g., arthrosis, osteochondritis dissecans, loose bodies inside the joint) (Fig. 8.1).

At first, the therapy is always conservative, and there are a variety of treatment options available: nonsteroidal anti-inflammatory drugs (NSAID), platelet-rich plasma (PRP) injections, physical therapy, manual therapy, extracorporeal shockwave therapy, acupuncture, and bandages. Often, only a combination of several methods is successful for treatment, since the complaints can be very persistent [7–9]. Eccentric strength train-



Fig. 8.1 MRI of an elbow with a medial epicondylitis humeri and a perifocal edema

ing, stretching measures, and antagonist training often relieve symptoms and should be integrated into the training routine to prevent injuries in the sense of adjunct compensatory training for climbers (ACT; e.g., www.act.clinic) [10]. In addition, circular elbow tape is recommended during climbing or sport-specific training (see also Chap. 22 Taping) [11]. If there is no alleviation of symptoms, surgical therapy should be considered at the earliest after 9–12 months once conservative treatment options have been exhausted. Fortunately, this only happens very rarely. In such cases, open and arthroscopic procedures with the aim of denervation of the respective epicondyle and excision of diseased tissue are available. Postoperatively, the respective arm is immobilized for 2 weeks. In addition, maximum stress should be avoided for 6 months postoperatively [7].

Lateral and Medial Epicondylitis

- **History:** Degenerative processes in the context of overload-related microtrauma and muscular imbalances
- **Onset:** Chronic
- **Clinical:** Pressure pain at the epicondyle, pain radiating into the forearm muscles, reduction in strength, and pain at rest
- **Diagnostic:** Medical history, clinical finding, sonography, MRI
- **Therapy:** Mostly conservative
- **Outcome:** Often long-lasting complaints

8.3 Tendinopathies of the Biceps, Triceps, and Brachialis Muscles

Although much rarer than epicondylitis humeri, tendinopathies of the biceps, triceps, and brachialis muscles can also be triggers for pain in the elbow region. As with epicondylitis, the cause of the complaints is a repetitive overuse of the muscle attachments with consecutive chronic microtraumas and muscular imbalances that ultimately lead to tendinosis. Clinically, there is an anterior elbow pain in the area of the distal biceps and

brachialis muscle tendon or a pain above the dorsal elbow in the area of the tendon insertion of the triceps muscle at the olecranon. As in epicondylitis, clinical examination and sonography are the main diagnostic tools. Further diagnostics by means of X-ray and MRI should only be carried out in the case of persistent complaints.

Tendinopathies around the elbow are also common in other sports, and their therapy is not significantly different for climbers. The therapy is conservative, whereby the reduction of stress as well as prevention by training the antagonist muscle groups has the highest priority. Eccentric exercises of the affected muscles are also an important part of the therapy. Similar to the muscular imbalances of the forearm flexors and extensors for epicondylitis, an imbalance between the pro- and supinator muscles of the forearm plays an essential role in the development of tendinopathies of the biceps and brachialis tendon. The biceps brachii muscle is the strongest supinator (outward rotation) in the elbow joint; the brachialis muscle is the strongest flexor in the elbow joint. The resulting torque is rather unfavorable when climbing, as the forearm is usually turned inward when gripping. Consequently, the sole pull of the biceps muscle on a horizontal hold would cause the hand to turn outward, and the hold could no longer be held. This overload can only be prevented by strongly trained pronator muscles (pronator teres, pronator quadratus, brachioradialis muscles) and flexor muscles (e.g., brachialis muscle). In addition, training pull-ups with complete extension in the elbow joint promote the development of a tendinopathy of the triceps tendon, which should therefore be avoided by means of a certain minimum residual flexion in the elbow joint [10, 11].

Tendinopathies of the Biceps, Triceps, and Brachialis Muscles

- **History:** Degenerative processes in the context of overload-related microtrauma and muscular imbalances
- **Onset:** Chronic
- **Clinical:** Pain around the elbow during load as well as local pressure-induced pain

- **Diagnostic:** Medical history, clinical finding, sonography, if applicable X-ray and MRI
- **Therapy:** Conservative
- **Outcome:** Often tedious complaints

8.4 Functional Compartment Syndrome of the Forearm Muscles

Load-dependent pain in the area of the forearm muscles during climbing is common. In particular, the flexor muscles in the deep compartment of the forearm are exposed to great stress. Their effectiveness is usually the limiting factor for climbing performance, especially on longer climbing routes. However, if the complaints are very intense and persist disproportionately strongly for 1 or 2 days after climbing, one must consider the rare so-called “functional compartment syndrome” of the forearm muscles [4]. Primarily, climbers describe a long-lasting feeling of increased pressure in the forearm flexor muscles. The cause is a stress-induced, excessive swelling of the musculature, especially of the forearm flexors within the muscle fascia, which encloses the so-called compartment. The increased tissue pressure leads to a consecutive impairment of the microcirculation with disrupted outflow of blood and lymph and, in case of a progressive, fulminant course, to disturbances to the neuromuscular functions. In extreme cases, irreversible damage (tissue necrosis, etc.) can occur [4].

This is often preceded by a viral infection, such as glandular fever (infectious mononucleosis) caused by the Epstein-Barr virus (kissing disease), although the exact connection is still unclear [12].

Clinically, during and after exertion while climbing, strong pain in the area of the forearm musculature can be found, going beyond the

familiar feeling of “thick forearms” and lasting longer. The pain is often characterized as “dull,” “as if the skin on the forearm was too tight.”

In advanced stages, paresthesia and circulatory disorders may occur. However, patients typically have no complaints without a corresponding prior exposure.

This rare pathology is difficult to diagnose. In addition to the medical history and the clinical examination, the diagnosis is confirmed by means of compartment pressure measurement under sport-specific stress. For this purpose, special probes are used, which are inserted into the respective compartments via cannulas (Fig. 8.2). In a study of ten competition climbers, reference values for physiological pressures could be determined by means of dynamic compartment measurements. At rest, the pressure should be below 15 mmHg; with sport-specific stress, the pressure can rise to values of up to approx. 30 mmHg. Within a recovery period of 15 min after exertion, pressures up to 15 mmHg are physiological. However, if pressures of 15–30 mmHg occur during the recovery period, there is a high risk of stress-induced functional forearm compartment syndrome. Moreover, pressures of over 30 mmHg confirm the diagnosis. In most cases, conservative therapy is sufficient (e.g., physical rest, stretching of the musculature, ice pads, and loosening massages). Nonsteroidal anti-inflammatory drugs (e.g., ibuprofen) can also have a decongestant effect. Despite these measures, if further deterioration of the findings occurs without further load with progressively increasing compartment pressures, the only remaining treatment option to avoid irreversible damage is a surgical fasciotomy of the forearm (Fig. 8.3). Fortunately, there are only few such climbing-related cases known in literature. Even in these dramatic cases, a complete recovery of the sport-specific performance capacity was achieved in the course of the treatment [4].



Fig. 8.2 Compartment pressure measurement



Fig. 8.3 Condition after fasciotomy of the forearm (compartment release) in a climber due to a rare but severe case of functional compartment syndrome

Functional Compartment Syndrome of the Forearm Muscles

- **History:** Swelling of the forearm musculature due to overstrain with consecutively restricted blood and lymph circulation
- **Onset:** Acute to chronic
- **Clinical:** Pain in the muscles of the forearm during and after exercise, possibly swelling
- **Diagnostic:** Medical history, clinical finding, sonography, compartment pressure measurement if necessary
- **Therapy:** Mostly conservative, fasciotomy (compartment release) only necessary in the most severe cases!
- **Outcome:** Good

8.5 Elbow Dislocation (with Vascular Injury)

Severe traumatic injuries, such as dislocations, sometimes even with vascular injuries around the elbow and the forearm, have been rare. Only a few case reports due to climbing accidents are described in literature [13, 14]. However, in the subdiscipline of bouldering, there has been an increasing incidence of elbow dislocations in recent years. This is most likely due to the low fall height with a backward fall vector onto a stretched or slightly flexed forearm. The diagnosis can usually be made clinically due to the malalignment and the often pain-induced complete loss of



Fig. 8.4 MRI after elbow dislocation with rupture of the ulnar collateral ligament

mobility in the elbow joint. X-ray imaging is required in order to exclude bony injuries.

The reduction should be performed under analog sedation or, if necessary, under anesthesia. The latter has the advantage that the ligament stability of the elbow joint can be tested. Subsequently, immobilization and additional diagnostics are performed by means of MRI (Fig. 8.4) and, in the case of concomitant bony injuries, by means of an additional CT scan to evaluate the extent of the pathology.

Surgery is the standard therapy in case of a tendency toward dislocation at flexion over 30° in the elbow joint or an instability in several examination areas (varus and valgus instability) due to multiple ligament injuries as well as dislocated fractures. The ruptured collateral ligaments are reattached with suture anchors (sometimes additional braced with FiberTape®), and bony injuries are addressed according to the guidelines of orthopedics and trauma surgery. Subsequently, the follow-up treatment consists of an early functional therapy using a movable elbow orthosis and treatment protocol, which is tailored to the extent of the injury.

Good functional results were achieved in climbers with a return to climbing after an average of 6 months [15]. Furthermore, rare second-

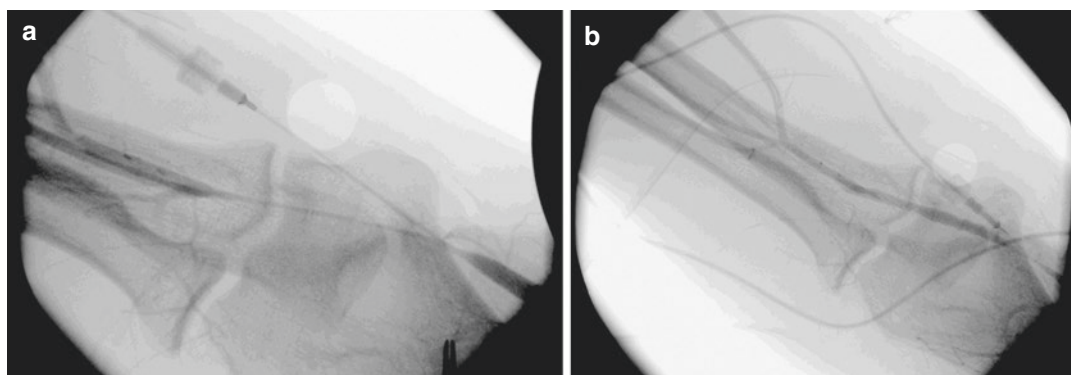


Fig. 8.5 Subtraction angiography depicting a lesion of the brachial artery following an elbow dislocation before (a) and after (b) patch plasty

any pathologies, such as vascular lesions, must be excluded or correctly treated within the diagnostic process. The following case study from our own patient population should provide increased awareness of this issue:

8.5.1 Elbow Dislocation with Injury to the Brachial Artery

A 24-year-old, otherwise healthy male climber had suffered a posterior, closed elbow dislocation during a fall while bouldering which was promptly reduced in analog sedation. Subsequently, the peripheral circulation, motor function, and sensitivity were without pathological findings.

The plain X-rays of the elbow showed a non-displaced fracture of the radial head. Afterward, the patient was discharged after immobilization of the respective arm with a plaster splint. In the further course of diagnostics, an MRI was performed, and the patient visited our clinic 2 weeks after the accident. During this consultation, peripheral circulation, motor function, and sensitivity were still intact. However, an ulnar-sided, strand-like induration with local pressure pain, which was initially interpreted as a hematoma after a brachialis muscle tear, was described in the antecubital fossa.

The MRI showed injuries of the medial and lateral collateral ligaments as well as a partial

rupture of the brachialis tendon with perifocal hematoma. Due to additional clinical instability, we decided on a surgical therapy. During the operation 5 days later, a long-distance stenosis of the brachial artery in the antecubital fossa and a thrombosis of the radial artery with just sufficient blood flow through collateral vessels were observed as a correlation to the previously palpated induration as a result of an injury to the brachial artery during the previous elbow dislocation (Fig. 8.5). Intraoperatively, an angiography and a subsequent thrombectomy and a venous patch plasty were performed. The ruptured ligaments were reattached, and all pathologies could be addressed. During follow-up 1 year after surgery, the patient was symptom-free even during intensive physical exertion [13].

Elbow Dislocation (with Vascular Injury)

- **History:** Traumatic
- **Onset:** Acute
- **Clinical:** Pain, misalignment, and impaired function
- **Diagnostic:** Medical history, clinical finding, (duplex-) sonography, MRI, and (MRI-) angiography if necessary
- **Therapy:** Reduction in analgesedation or anesthesia, surgery dependent on the extent of the pathology (e.g., instability, bony injuries)
- **Outcome:** Good

8.6 Muscle Injuries

Severe, traumatic muscle injuries in the area of the elbow and forearm as a result of accidents while climbing and bouldering are fortunately a rarity. Nevertheless, these must be excluded in the diagnostic process to avoid irreversible consequences. The following case study is intended to sensitize the attending physician to those rare but severe injuries:

8.7 Rupture of the Short Head of the Biceps Brachii Muscle Belly

A 42-year-old, otherwise healthy male climber fell while climbing in a bouldering gym.

He was climbing around a volume when his foot slipped. Consequently, his right upper arm hit the edge of the volume while the biceps mus-

cle was fully tensed, a blunt trauma that resulted in a rupture of the short head biceps muscle belly. This caused immediate pain, swelling, and a palpable gap in the midportion of the upper right arm. The X-ray and MRI images showed a complete, isolated rupture of the short head of the biceps brachii muscle but no fracture (Fig. 8.6). In accordance with recent literature, we restored the anatomical integrity of the muscle belly itself through an open surgical approach with direct end-to-end modified Mason-Allen stitches, 10 days after the trauma, to prevent the ruptured muscle belly from further retraction, which would later hinder anatomical reduction (Fig. 8.7) [16]. As such, our patient was able to regain full strength and functionality. Six months after surgery, he had already returned to a climbing level close to his pre-injury level. Furthermore, the clinical investigation showed a full range of motion without functional complaints or pain [14]. In general, the majority of muscle injuries

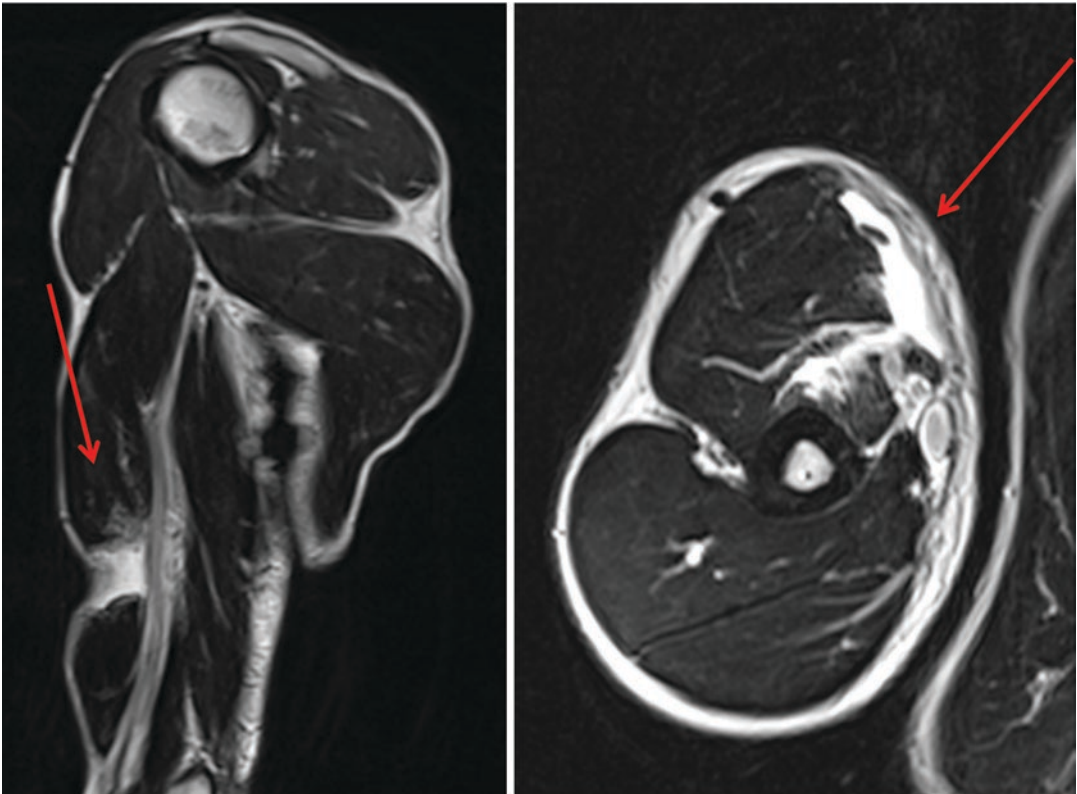


Fig. 8.6 MRI of an upper arm depicting a rupture of the short head of the biceps brachii muscle belly

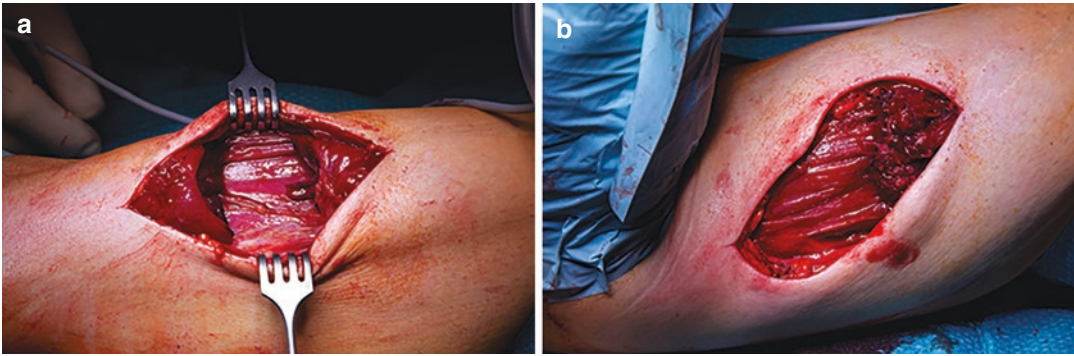


Fig. 8.7 Intraoperative clinical findings of a rupture of the short head of the biceps brachii muscle belly before (a) and after (b) direct end-to-end suture

can be managed conservatively with measures such as rest, nonsteroidal anti-inflammatory drugs, and physical therapy. There are only a few indications for surgery: a large intramuscular hematoma, tears of more than half of the muscle belly (Grade 3b-4), and the occurrence of a compartment syndrome, which is an absolute emergency [17].

Muscle Injuries

- **History:** Traumatic
- **Onset:** Acute
- **Clinical:** Pain, limitation of strength and function, in some cases palpable gap
- **Diagnostic:** Medical history, clinical finding, sonography, MRI if necessary
- **Therapy:** Conservative for partial muscle ruptures and strains, surgical for subtotal and complete muscle ruptures
- **Outcome:** Good

8.8 Distal Biceps Tendon Rupture

Ruptures of the distal biceps tendon account for only 3% of all tendon injuries of the biceps brachii muscle. However, due to the resulting loss of strength, especially in climbers, it requires a precise diagnosis and therapy. In most cases, an avulsion of the tendon attachment at the tuberosity radii can be found, which occurs mainly in men between 30 and 60 years of age as a conse-

quence of chronic, degenerative changes. Additional risk factors are regular nicotine consumption, excessive strength training, and abuse of anabolic steroids.

Most often, the injury mechanism is described as a sudden force on the flexed and supinated forearm by pulling on an undercling into a “lock-off” position followed by a clicking sound [11].

This is followed by pain in the antecubital fossa and an immediate loss of strength, especially for supination but also in flexion of the elbow joint. The diagnosis can usually be made clinically. In addition to the previously mentioned complaints, the distal biceps tendon can no longer be palpated (hook test), and a so-called “reversed Popeye sign” (the biceps brachii muscle belly retracts proximally) can be seen in the case of a complete rupture. The diagnosis is confirmed by sonography and X-ray. In case of ambiguous findings, such as partial ruptures or bony lesions, MRI imaging is recommended (Fig. 8.8). Since conservative therapy can reduce the force for supination movements by up to 80%, prompt surgical therapy is the gold standard, especially for athletes (even partial ruptures of more than 50%). Thereby, the torn tendon is reattached to its anatomical insertion point on the radius using suture anchors. An early reattachment is essential to prevent tendon retraction and the development of muscle atrophy and fibrosis [18].

Subsequently, a partial immobilization of the elbow joint is necessary for 6 weeks, which is gradually reduced under physical therapy

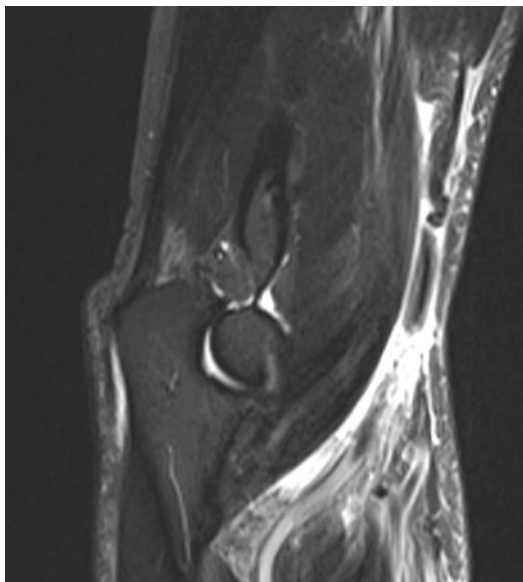


Fig. 8.8 MRI of an elbow depicting a distal biceps brachii tendon rupture

guidance (initial plaster splint, followed by elbow joint orthosis with reduced range of motion) [18]. Regarding the postoperative outcome, a recent review by Thomas and Lawton shows excellent results in athletes with a complete recovery of athletic performance and a full range of motion without functional complaints or pain [19].

Distal Biceps Tendon Rupture

- **History:** Traumatic (mostly in case of preexisting degenerative changes)
- **Onset:** Acute
- **Clinical:** Pain in the elbow and lack of strength for flexion and especially supination in the elbow joint
- **Diagnostic:** Medical history, clinical finding, sonography, X-ray, MRI if necessary
- **Therapy:** Surgical
- **Outcome:** Good

8.9 Conclusion

Most of the pathologies around the elbow are caused by overstrain (e.g., medial and lateral epicondylitis humeri, tendinopathies of the biceps,

triceps, and brachialis muscle). Their therapy is almost always conservative. Since training errors are often the trigger for the development of the complaints, prophylaxis through adjunct compensatory training plays an essential role. Traumatic injuries such as distal biceps tendon ruptures as well as rare pathologies such as functional compartment syndrome of the forearm or traumatic vascular and muscle injuries must also be taken into account as a differential diagnosis.

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